Aerospace Structures Design (AerE 4260/5260)

Final Project Report

The Swash-Bucklers

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1. Introduction

A. Design Requirements

- Geometric and Structural Requirements

- Beam Depth:
  - Maximum Depth: 10.5 inches for notched seat track design
- Support and Attachment points:
  - Beam attaches to a shear-tied zee frame
  - Beam Supported by a square tube stanchion with 1.8 x 1.8 inch outside dimension and 0.080-inch wall thickness (7075-T651 alloy)

- Load Requirements
  - Ultimate Loads:
    - Safety factor: 1.5
    - Emergency landing loads, decompression loads, and flight loads specified as ultimate loads without the safety factor.
    - Vertical seat track attachment loads:
      - 4,000 lbs at BL 33.99 and BL 75.92.
      - 6,000 lbs at BL 11.33 and BL 98.58.
  - Stiffness:
    - Vertical Deflection must not exceed 1.0 inch

- Fatigue and Durability
  - Fatigue Life:
    - Must be designed for 20,000 flight cycles
    - Minimum load assumed to be zero
  - Damage Tolerance:
    - Designed to comply with FAR 25.571 for damage tolerance.
  - Corrosion Resistance:
    - Design for mildly corrosive environment.

- Material and Construction
  - Material Options:
    - Aluminum alloys: 7075-T651 or 2024
    - Composite materials like Carbon Fiber Reinforced Plastic
- Structural Stability
  - Stabilization Straps (if required)
    - Sized using 5% of the summation of ultimate compressive load in all chords for tensile load path.
  - Beam Joint Attachments
    - Fastener Options: aluminum rivets or titanium bolts
    - Typical fastener spacing: 4D to 6D, with edge margins of 2D (D = fastener diameter).

- Structural Analysis and Validation
  - Load analysis
    - Includes deflection, bending moments, shear, buckling, and joint stresses.

- Validation
  - The final beam must demonstrate compliance with all ultimate loads, stiffness, fatigue, and damage tolerance requirements.

B. Choice of Material

For the design, aluminum 7075-T651 is chosen. Aluminum is significantly cheaper and easier to manufacture than carbon fiber-reinforced plastic. It also performs under a broader range of temperatures and has a higher fatigue resistance and durability compared to CFRP. Due to the choice of bolted joints, aluminum is more optimal for the design, as there is difficulty drilling holes into CFRP. For these reasons, all components with the exception of rivets will be aluminum 7075-T651.

C. Floor Beam Design

Our design cross-section properties and drawing are shown in Section 2.D. Our beam follows the design constraints of the problem. It has 11 stiffeners placed 20.5 in apart from each other, with a central beam located at BL 0. Besides, four holes with 5 in diameter were placed at LBL 30.75 in and LBL 71.75 in, the other two in in symmetric locations on the right side.

The beam was also modeled in ANSYS, with the stated applied loads and constraints, as seen in the figure bellow:

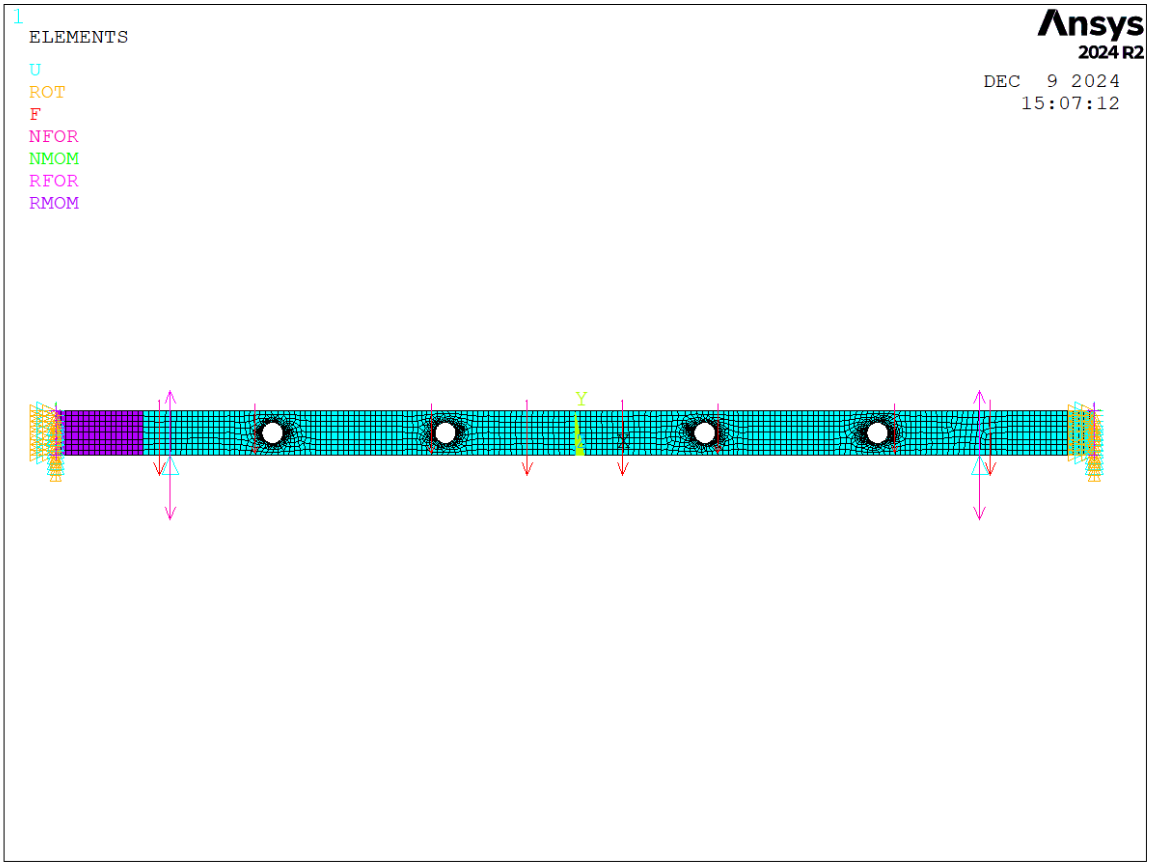

Figure 1. Floor beam geometry (meshed) and applied loads and constraints.

D. Parts List

Part	Material	Size	Quantity	Price
I-Beam	Aluminum 7075-T651	20.5 ft	1	~ $3,500
Stiffeners	Aluminum 7075-T651	1 x 1 in T = 0.188 in	22	~ $120
Stiffener Rivet	A-286 CRES	¼ inch diameter	110	~ $15 for pack of 100
Floor Chair Bolts	Titanium SAE Grade 5	¼ inch diameter	16	~ $64 for 16
Stanchion	Aluminum 7075-T651	1.8 X 1.8 in	2	~ $200
Stanchion Rivets	Aluminum 7075-T73	3/8 inch diameter	14	~ $110 for pack of 100
Z-Frame	Aluminum 7075-T651	4.12 in	2	~ $180
Z-Frame Rivets	Aluminum 7075-T73	5/16 inch diameter	8	Use same rivets as Stanchion
Total Cost				~ $4,189

2. Floor Beam – Structural Analysis

A. Load paths

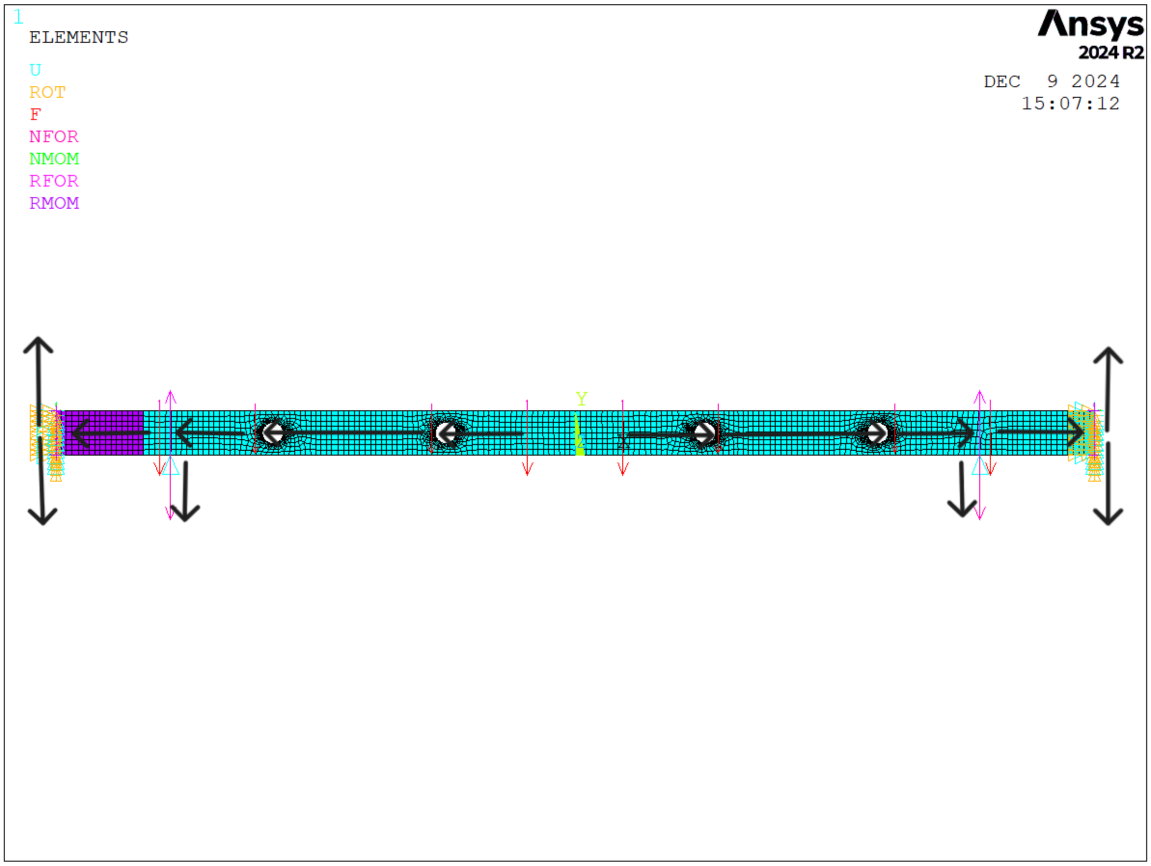

Figure 2. Floor beam load paths (black arrows).

B. Shear and bending moments diagrams

From the assignment sheet, the axial, shear and moment loads for the symmetric half-section are given as:

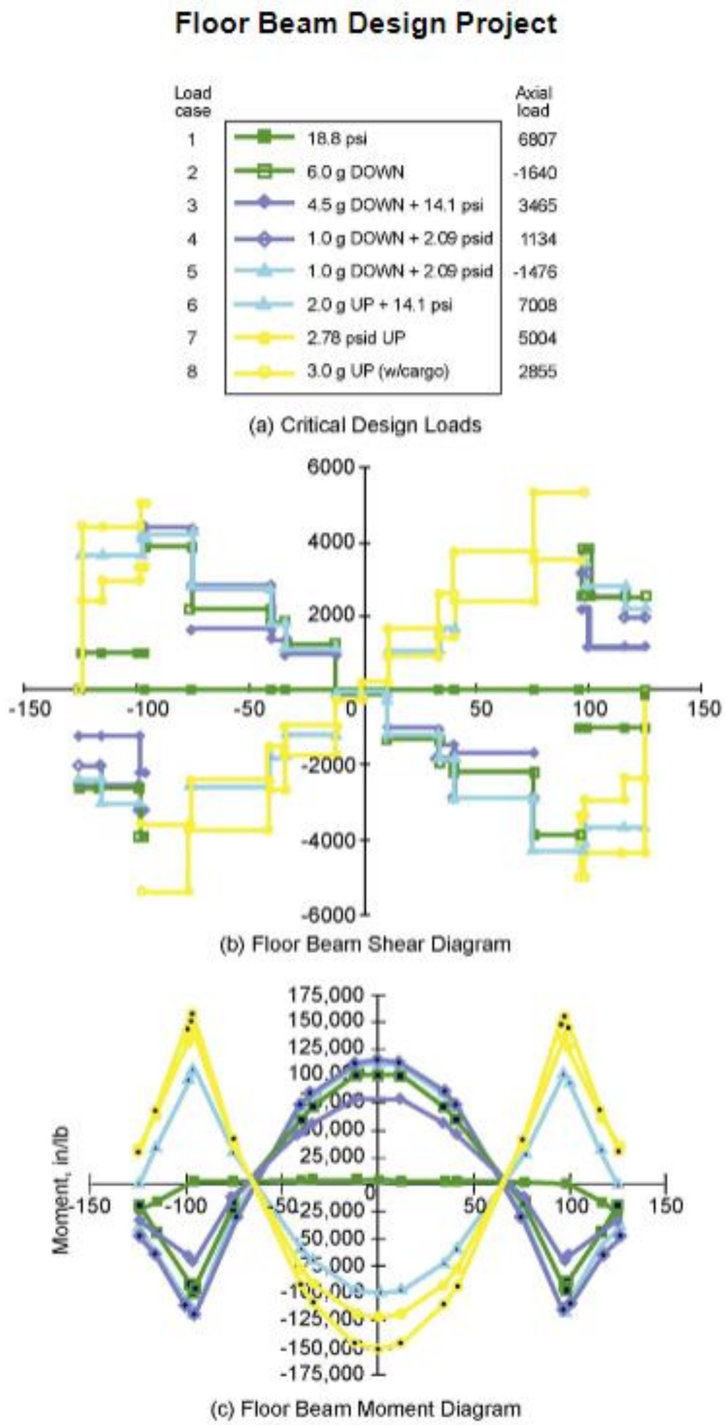

Figure 5. Station 1600 Floor Beam Shear and Moment Diagram

Figure 3. Critical Design Loads table, Floor Beam Shear Diagram, and Floor Beam Moment Diagram.

C. Ultimate-strength check for floor beam details

Since yield strength has been used as the primary load criterion for the part list (no brittle materials). Parts being safe under yield implies that they are safe under ultimate strength.

D. Section properties

The beam design is an I-Beam with the following cross-sectional dimensions:

Flange Thickness = 1 in
Web/Skin Thickness = 2 in
Width (b) = 6 in
Height (h) = 10.5 in

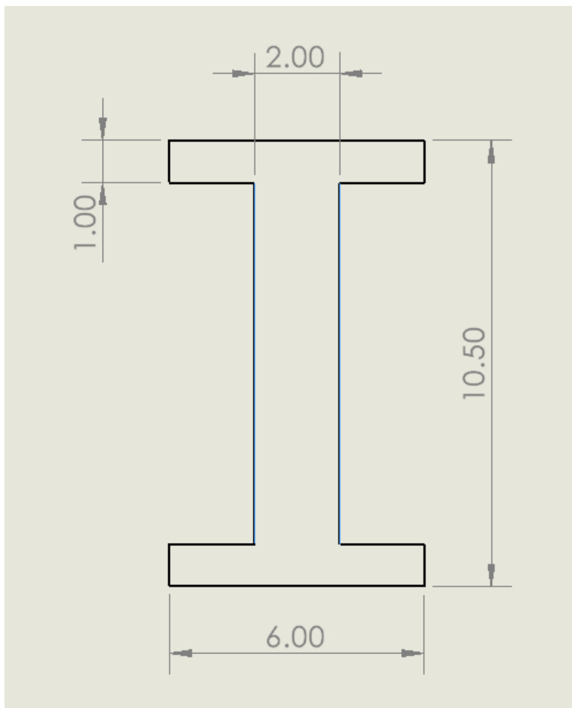

Figure 4. Beam cross-section dimensions in inches.

E. Stabilization

The maximum axial compressive force that the beam experiences is 1,640 lbs (from (a) Critical Design Loads table). Using ANSYS to model and analyze the beam, we got a buckling factor of 3.97 from the first buckling mode. Thus, the critical buckling load is 6,510.8 lbs. Additionally, we then know the factor of safety to be 3.97, and therefore the beam is stable under the load conditions, no straps are required on the beam.

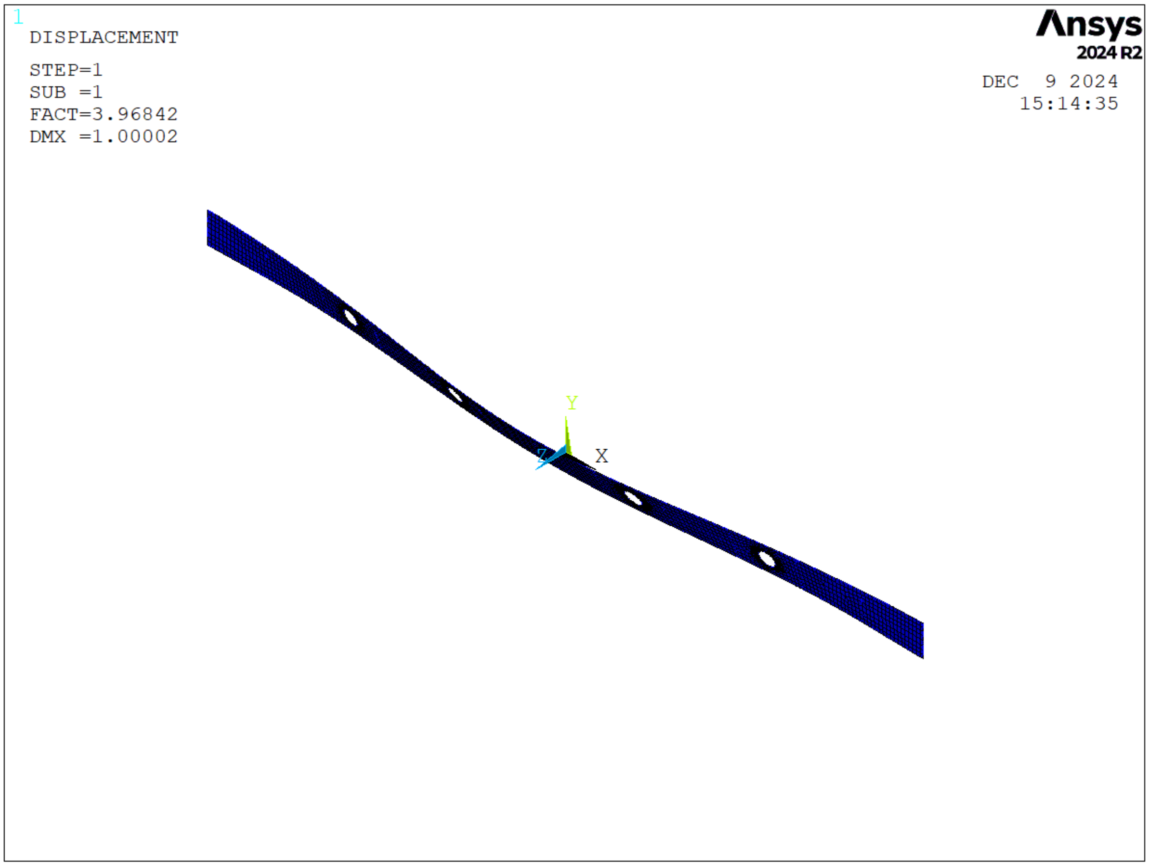

Figure 5. Floor beam mode 1 buckling.

F. Maximum vertical deflection

From the same analysis using ANSYS, the maximum vertical deflection of the beam under the stated load and constraints conditions is 0.13 in downwards.

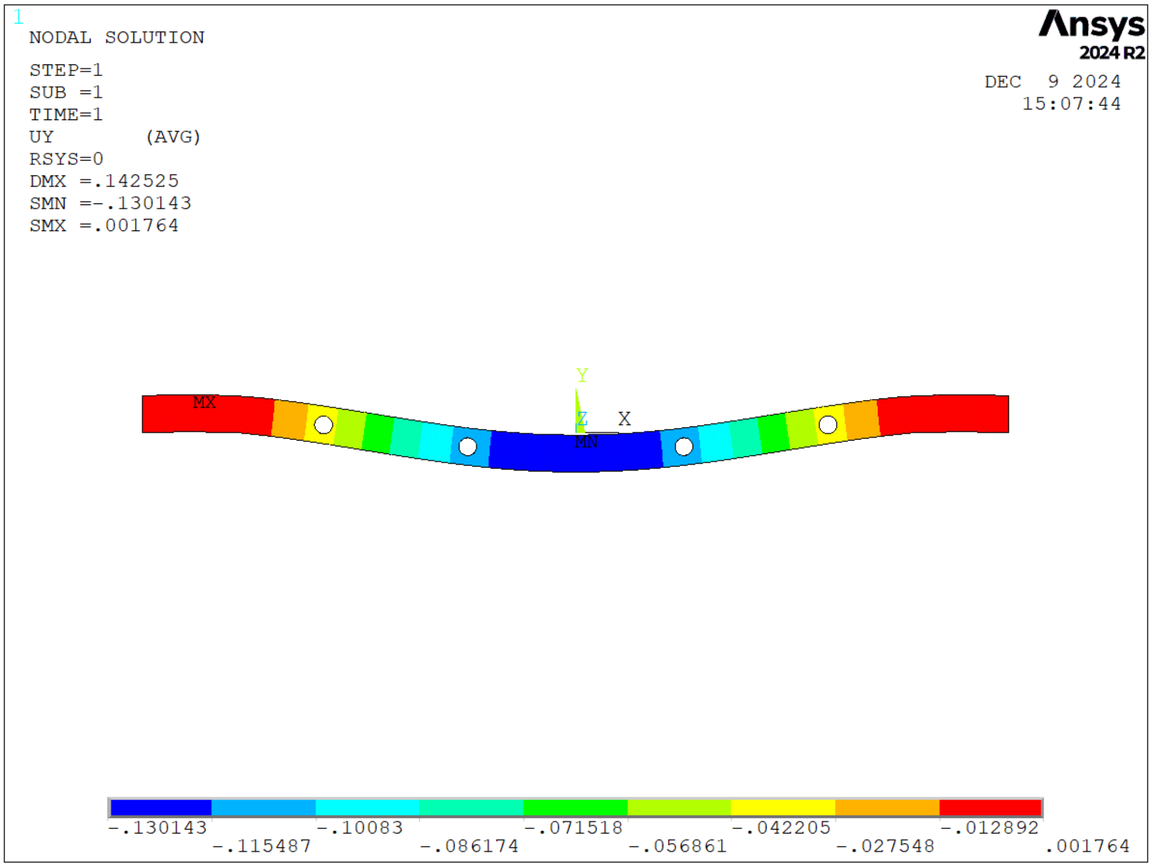

Figure 6. Floor beam vertical (Y) displacement contour plot.

G. Stiffeners and Stiffener-Web Rivets

The material used is for the stiffeners is aluminum 7075-T651. Let the spacing between each stiffener be 20.5 inches, which is corresponds to 11 stiffeners across the entire beam. The compressive force on a stiffener can then be calculated with a factor of safety of 1.5 and $\alpha = 42.5°$:

$$P = 1.5 \cdot V \frac{w}{h} \tan(\alpha)$$

$$P = (30{,}000) \frac{20.5}{10.5} \tan(42.5)$$

$$P = 53{,}670.8\ lbs$$

To obtain an initial value for moment of inertia, the equation for critical compression load can be used:

$$P_{cr} = \frac{\pi^2 E I_{st}}{L^2}$$

$$53{,}670.8 = \frac{\pi^2(10.6 \cdot 10^6)I_{st}}{10.5^2}$$

$$I_{st} = 0.0566\ in^4$$

To select an item from the manufacturer, this value must be halved, as loads will be distributed over 2 parts. Item #2 from the manufacturers list given in Lab 4 will be selected due to its moment of inertia and cross-sectional area. The following are the characteristics of this item:
Ixx=0.0649 in^4
Area=0.376 in^2

Then the critical buckling load can be calculated for this item:

$$P_{cr} = 123{,}169\ lbs$$

The actual compression load on the stiffener:

$$P = (20{,}000)\frac{20.5}{10.5}\tan(42.5) = 35{,}780.6\ lbs$$

The resulting factor of safety for buckling is 3.44. The compressive stress on each stiffener is as follows:

$\sigma = \frac{P}{A}$

$$\sigma = \frac{35{,}780.6}{2 \cdot 0.376} = 47.58\ ksi$$

With aluminum 7075-T6 having a compressive yield of 71.7 ksi, the resulting factor of safety is 1.51.

For rivet design the following equations are used:

Vertical Rivet Load Transfer – **qcosec(α)**

$$q = \tau * t$$

$$\tau = V/ht$$

Where t is the thickness of the web, 2 inches. This results in a vertical rivet load transfer of 4229.1 lbs/inch. Due to this load, the ¼ inch A-286 CRES rivet will be chosen, with an ultimate single shear load of 4415 lb/s fastener. The rivet spacing is then calculated as follows:

$$\frac{4415\ \frac{lb}{fastener}}{4229.1\ \frac{lb}{inch}} = 1.04\ \frac{inches}{fastener}$$

This results in 10 rivets for each stiffener-web attachment. Below is the cross-section of Item #2. There are two for each stiffener – 22 in total. Each has a length of 10.5 inches.

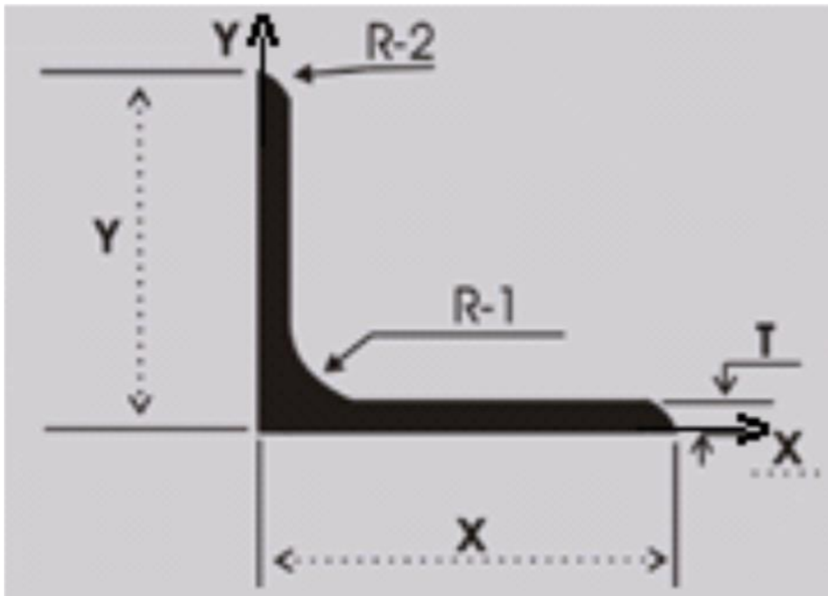

Figure 7. Stiffeners cross-section.

Y=1 in X=1 in T=0.188 in
R-1=0.125 in R-2=0.094 in

H. Stress Analysis

The following analysis uses reaction forces found in ANSYS. These reaction forces on the stanchion and frame are given below. Nodes 1 and 58 indicate the frame while nodes 10 and 47 indicate the stanchions.

PRRSOL Command

File

```
PRINT F    REACTION SOLUTIONS PER NODE

 ***** POST1 TOTAL REACTION SOLUTION LISTING *****

 LOAD STEP=     1  SUBSTEP=     1
  TIME=    1.0000      LOAD CASE=   0

 THE FOLLOWING X,Y,Z SOLUTIONS ARE IN THE GLOBAL COORDINATE SYSTEM

   NODE       FX            FY           FZ
      1   7008.0      0.10738E-021 -221.30
     10                            -19779.
     47                            -19779.
     58  -7008.0     -0.10738E-021 -221.30

TOTAL VALUES
VALUE    0.0000     -0.61196E-034 -40000.
```

Figure 8. Reaction forces solution per node.

i. Floor-beam-to-frame attachment

The floor beam is attached to the fuselage through a z-frame as shown below (from the assignment document):

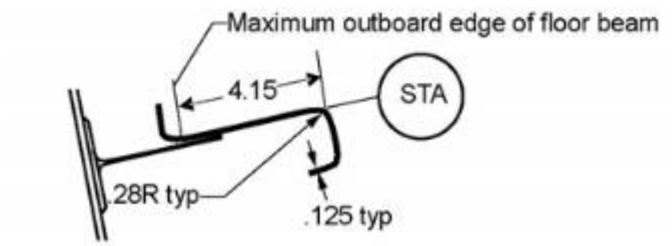

Figure 9. Z-frame fuselage attachment.

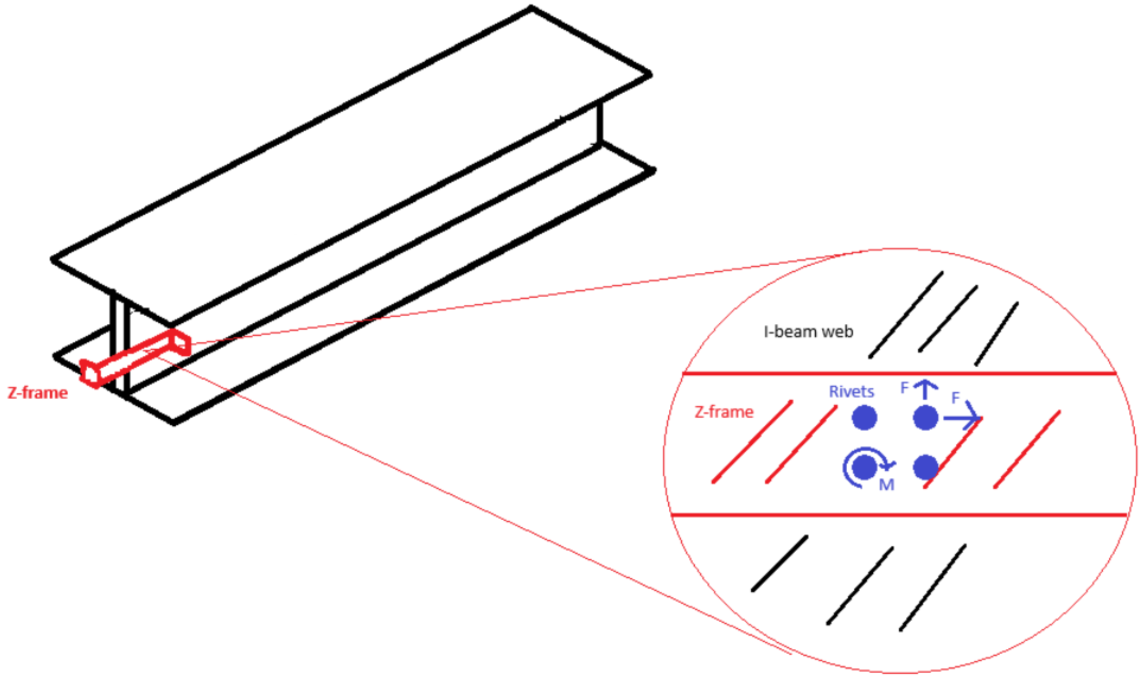

Figure 10. Beam and Z-frame attachment; Zoom-in rivets configuration.

In order to design the rivets for the frame, the force on each rivet must be calculated. In order to achieve this, firstly a coordinate system must be established. The (0,0) was chosen to be the bottom left corner of the attachment. From here, a square formation of four 7075 aluminum rivets was tried. The centroid of the formation was found using the following equation.

$$\overline{X} = \frac{\Sigma DX}{\Sigma D}$$

$$\overline{Y} = \frac{\Sigma DY}{\Sigma D}$$

From here, the centroid location was determined to be (2.25,1.5). After this, the moment was able to be determined using the following equation and the forces Py of 221.3 lbs. and Px of 7,008 lbs. These forces are located at the centroid of the square formation.

$$M = \Sigma(P \cdot d)$$

Where d is the distance to the centroid. Because the distance to the centroid is zero for both force cases, the moment can be ignored. From here, the moment of inertia can be calculated using the following equation.

$$I = \Sigma DX^2 + \Sigma DY^2 - \overline{X}\Sigma DX - \overline{Y}\Sigma DY$$

Using this equation the moment of inertia was determined to be 0.8125. From here, the force per rivet was found with the following equation.

$$P_i = D\sqrt{\left(\frac{P_y}{\Sigma D} + \frac{M(X_i - \overline{X})}{I}\right)^2 + \left(\frac{P_x}{\Sigma D} + \frac{M(Y_i - \overline{Y})}{I}\right)^2}$$

Shear Analysis

The maximum load per rivet was determined to be 1,753 lbs. This led to a rivet diameter of 5/16 inches with a maximum load of 3,520 lbs. Being chosen. The factor of safety can be determined using the following equation.

$$Factor\ of\ Safety = \frac{Maximum\ Load\ per\ Rivet}{Actual\ Load\ per\ Rivet}$$

Using this equation, the factor of safety was determined to be 2.01

Bearing Analysis

The maximum load per rivet, determined to be 1,753 lbs. Can be compared to the bearing load of the plate. The bearing yield of the aluminum frame was found to be 110 ksi. Compared to the table, for 5/16 inch diameter rivets the maximum bearing load is 4,038 for a bearing yield of 100 ksi. At a thickness of 0.125 inches. Multiplying by the difference in the bearing yield gives a load maximum of 4,442 lbs. The load per rivet is 1,753 lbs. meaning that the frame will not fail due to bearing. The factor of safety found using the same method as above was determined to be 2.212.

ii. Floor-beam-to-stanchion attachment

The stanchion is attached to the floor beam using 7 aluminum 7075 rivets with a diameter of 3/8 inches. The riveting formation is shown below.

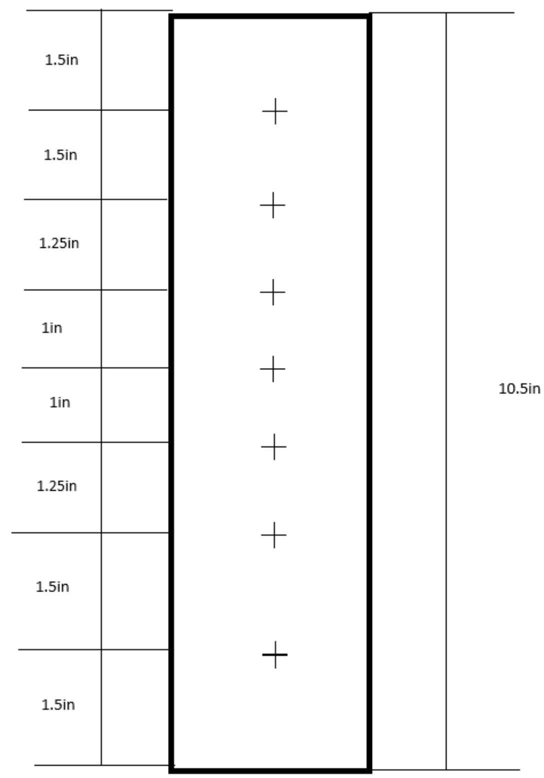

Figure 11. Stanchion to beam attachment rivets formation.

The following method was used to analyze the beam for both bearing and shear cases. Firstly, the coordinate points of the rivet centers were found, assuming the load is applied on the center of the beam vertically. This force was found in ANSYS to be 19,779 lbs. The arbitrary axis for analysis can be set to align with this force, setting (0,0) at the bottom middle of the diagram. After these coordinates were determined, the Centroid was found using the following equation.

$$\overline{X} = \frac{\Sigma DX}{\Sigma D}$$

$$\overline{Y} = \frac{\Sigma DY}{\Sigma D}$$

The centroid was found to be located at (0,5.25). From here, the moment must be considered. This moment can be neglected because it aligns with the centroid, because the moment arm is zero, the moment becomes zero. From this point, the moment of inertia can be found using the following equation.

$$I = \Sigma DX^2 + \Sigma DY^2 - \overline{X}\Sigma DX - \overline{Y}\Sigma DY$$

The moment of inertia was found to be 15.0938 in^4. From here the load on each rivet was found using the following equation.

$$P_i = D\sqrt{\left(\frac{P_y}{\Sigma D} + \frac{M(X_i - \overline{X})}{I}\right)^2 + \left(\frac{P_x}{\Sigma D} + \frac{M(Y_i - \overline{Y})}{I}\right)^2}$$

Shear Analysis

The maximum load per rivet was determined to be 2,879 lbs. This is within the acceptable limit of 5,030 lbs. For the chosen rivet. Calculating a factor of safety using the following equation, it was found to be 1.747.

$$Factor\ of\ Safety\ = \frac{Maximim\ Load\ per\ Rivet}{Actual\ Load\ per\ Rivet}$$

Bearing Analysis

The maximum load per rivet, determined to be 2,879 lbs. Can be compared to the bearing load of the plate. The bearing yield of the aluminum stanchion was found to be 110 ksi. Compared to the table, for 3/8 inch diameter rivets the maximum bearing load is 3,088 for a bearing yield of 100 ksi. At a thickness of 0.080 inches. Multiplying by the difference in the bearing yield gives a load maximum of 3,397 lbs. The load per rivet is 2,879 lbs. meaning that the stanchion will not fail due to bearing. The factor of safety found using the same method as above was determined to be 1.180.

iii. Floor-beam-to-continuous-seat-track attachment

The seat track attachments will be secured to the top of the floor beam using 4 SAE Grade 5, ¼ inch diameter titanium bolts with a maximum stress of 140 ksi. The stress on each bolt can be calculated using the following equation.

$$Stress\ Per\ Bolt\ = \frac{P}{4 \cdot \pi \cdot \left(\frac{d}{2}\right)^2}$$

Using a maximum load P of 6,000 lbs. And a diameter of ¼ inches, the stress per bolt was found to be 30,557 psi. This can be compared to the maximum stress of each bolt using the factor of safety. The following equation can be used to calculate a factor of safety for these bolts.

$$Factor\ of\ Safety\ = \frac{Maximum\ Stress\ per\ Bolt}{Actual\ Stress\ per\ Bolt}$$

Using this equation, the factor of safety was found to be 4.58.

I. Fatigue analysis

i. Damage tolerance

The fatigue load spectrum on the beam can be represented with an equivalent ground-air-ground (GAG) cycle:

$$F_{max} = 9.4psi + 1.3g\ down,\ F_{min} = 0$$

These load cases can be found from Figure 6 in the project instructions. The 9.4 psi load case can be scaled down from the 18.8 psi load case in Figure 6a and the 1.3g down load case can be scaled down from the 4.5g load case in Figure 6b. Then the stress from the moments and axial loads can be superimposed to obtain the maximum stress for the analyzed load case.

$$M_{max} = \max\left(abs\left(M_{max}^{g} + M_{max}^{pressure}\right)\right)$$

$$F_{max} = \frac{M_{max}c}{I} + \frac{P_g + P_{pressure}}{A}$$

One the maximum stress is found, the stress ratio can be calculated as follows.

And the stress ratio $R = \frac{F_{min}}{F_{max}}$

Then R and Fmax can be used to look up the number of cycles to failure in an S-N curve. An S-N curve is provided for Aluminum 7150-T77511 in Figure 11 of the project assignment:

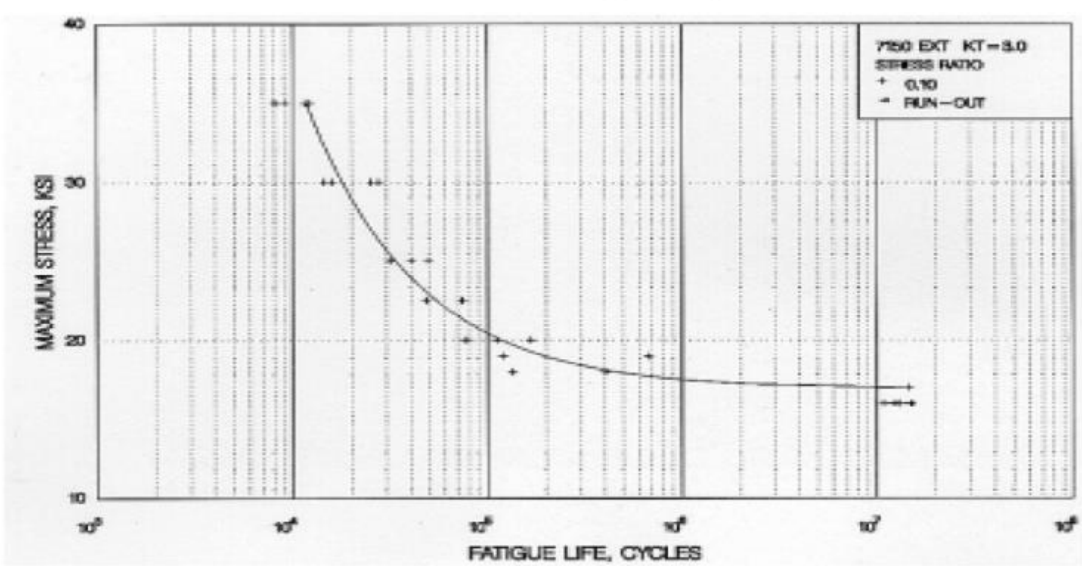

Figure 12. S-N curve for Aluminum 7150-T77511.

The maximum stress along with the R-ratio (which will be zero for the load case that is being analyzed) can be used to find the number of fatigue life cycles for the floor beam. The factor of safety can be calculated as

$$FS_{fatigue} = \frac{N_{required}}{N_{actual}}$$

from the required number of life cycles from the project along with the actual calculated number of life cycles to failure.

For the I-beam section properties listed in Section D, the maximum stress was evaluated in MATLAB to be Fmax = 550.490 psi. This is below the endurance limit shown in the S-N curve above, so this particular beam section is safe from fatigue in the given load case.

ii. Fatigue evaluation of the critical system penetrations in the web of the floor beam

While the holes used for critical system penetrations are stress concentrators, the largest stress concentrations are at the locations where the stanchions attach. Shown below is a distribution of Von Mises stresses throughout the beam simulated in ANSYS. The color bar represents Von Mises stresses in units of psi:

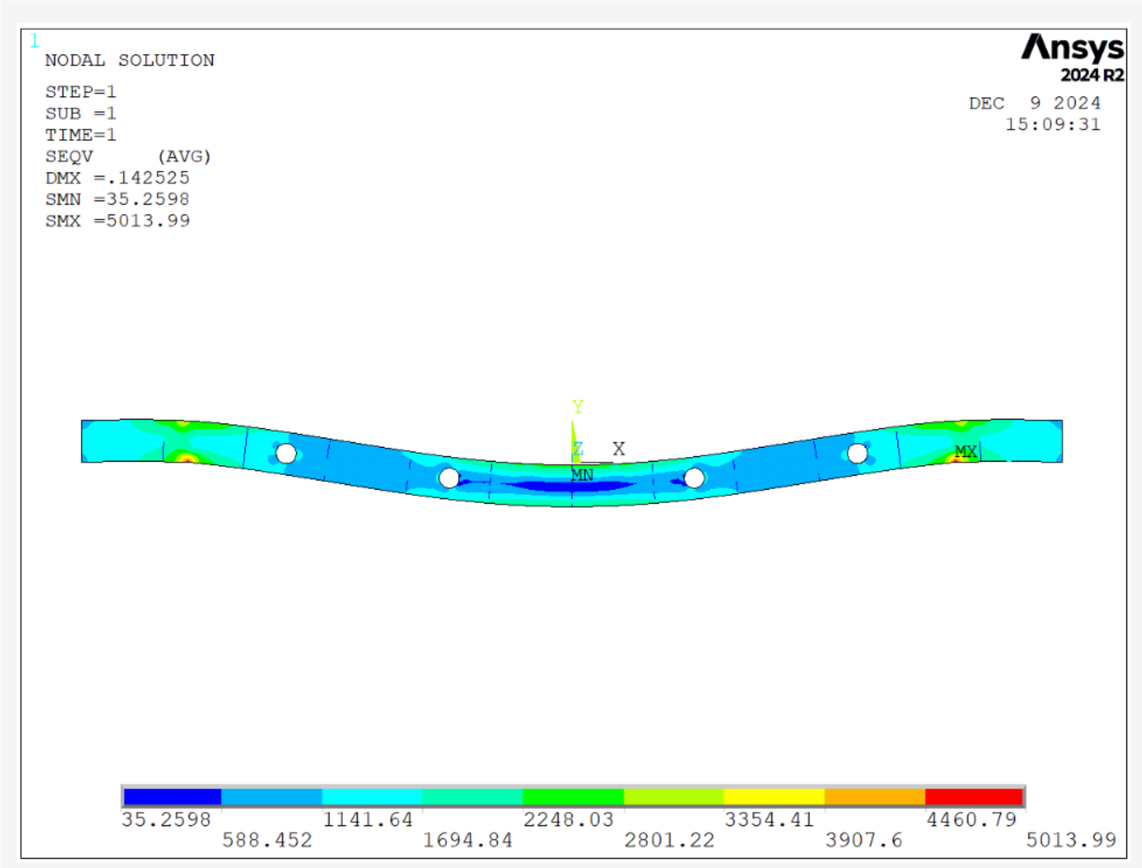

Figure 13. Floor beam Von Mises stress contour plot.

The largest stresses in the beam will then be 5.014 ksi, which are also below the endurance limit for the figure in Section J.i.

iii. Margin of safety calculation

The margin of safety is described as: $MS = \frac{F_{max}}{f_{max}} - 1$

Where Fmax is the allowable max stress required to achieve the goal number of time-of-life (TOL) cycles, and fmax is the actual maximum operating stress. The largest stress concentration will come from the stanchions, so the maximum stress from the stanchions will be used for maximum operating stress (fmax) in the margin of safety calculation. From the figure in Section J.i., the minimum number of life cycles is 20,000. Assuming that the minimum stress is still zero (R = 0), then Fmax = 29 ksi. Then the margin of safety will be:

$$MS = \frac{29\ ksi}{5.014\ ksi} - 1 = 4.784$$

J. Corrosion prevention

Since the floor beam being designed is not beneath a part of the plane that produces corrosive environments like a lavatory or galley, the environment the beam is subjected to will be considered as mildly corrosive.

For metals like Aluminum, it is generally best to apply a protective coating to mitigate corrosion. From Lecture 4 Page 6, 2XXX and 7XXX aluminum alloys are susceptible to stress corrosion cracking and exfoliation corrosion. Common methods for protecting aluminum surfaces are anodization, where a strong passive Aluminum Oxide (Al_2O_3) layer is created through the use of electrical current and additives ("Anodizing"). Protective paints and powders can also be used, for example Zinc Chromate is a pigment that can be used in Aluminum paints (Edwards and Wray). Aluminum is also known to undergo galvanic corrosion when in contact with copper (Selwyn), so any wires that run through the holes in the beam should be regularly inspected to ensure that there is a barrier between the wires and the beam.

Modern Carbon Fiber Reinforced Plastics (CFRPs) are made of polymers, which are subject to degradation from UV radiation due to UV wavelengths resonating with single Carbon-Carbon bonds. However, since the beam is inside the plane, this is less of a concern. However, some consideration may need to be made for protecting the connections between the beam and the fuselage, as CFRPs can be subject to galvanic corrosion with Aluminum (Yari). This could either be mitigated by making the fuselage out of CFRPs, by adding the aforementioned corrosion protections to the interior surfaces of the Aluminum fuselage, or by adding a physical barrier between the aluminum and the fibers like fiberglass (Composite Envisions).

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